***Electronic Appendix to the Manuscript "Metabolic costs of brain size evolution" by Karin Isler and Carel P. van Schaik***

## Appendix A: Brain mass data

Data on brain mass and body mass for 313 mammalian species were taken from unpublished data bases of R.D. Martin and G.M. Mace (after a shift of 0.59 grams in the rodents data from Mace et al. (1981) was corrected: G. Mace, pers. com.) as well as from Crile & Quiring (1940), Oboussier & Schliemann (1966), Mace et al. (1981), Stephan et al. (1981), Mace & Eisenberg (1982), Bauchot (1985), Gittleman (1986), Bernard & Nurton (1993), Smith & Jungers (1997), Iwaniuk (2001), Hutcheon et al. (2002), Mann & Towe (2003), Jones & MacLarnon (2004) and Kirk (2006). Cranial capacity of three species was measured directly on skulls using millet-seeds: *Manis javanica* (n=1), *M. tricuspis* (n=1) and *Orycteropus afer* (n=1).

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## Appendix B: Phylogeny

Phylogenetic relationships among orders are those proposed by Murphy et al. (2001b). Phylogenetic relationships within orders are as follows:

***Afrosoricidea and Eulipothyphla.*** We used the phylogeny proposed by Grenyer & Purvis (2003) completed with data from Fumagalli et al. (1999, Soricinae) and Quérouil et al. (2001, Crocidurinae). Note that we treated the insectivores as two separate orders (see Springer et al., 1997; Stanhope et al., 1998; Mouchaty et al., 2000; Murphy et al., 2001a; Murphy et al., 2001b).

***Artiodactyla.*** Relationships among infraorders and families are those proposed by Murphy et al. (2001a). The relationships within families follow Randi et al. (1998, Cervidae) and Matthee & Davis (2001, Bovidae) and are complemented with data from Hernández Fernández & Vrba (2005).

***Carnivora.*** We used the phylogenetic tree from Bininda-Emonds et al. (1999) and Flynn et al. (2005), complemented with phylogenies proposed by Yoder et al. (2003, Herpestidae) and Zrzavy & Ricankova (2004, Canidae).

***Chiroptera.*** Phylogenetic relationships are those proposed by Jones et al. (2002).

***Lagomorpha.*** Phylogenetic relationships are those proposed by Stoner et al. (2003).

***Primates.*** We used the phylogeny proposed by Purvis (1995), complemented with data from Canavez et al. (Canavez et al., 1999, Platyrrhini); and Roos et al. (2004, Strepsirrhini)

***Rodentia.*** Relationships among families are those proposed by Adkins et al. (2001), Huchon and Douzery (2001) and Montgelard et al. (2002); relationships within the Cavioidea follow Rowe and Honeycutt (2002). Relationships within families are from Walton et al. (2000, Bathyergidae); Hafner et al. (1994, Geomyidae); Hafner & Hafner (1983, Heteromyidae); Michaux et al. (2001, Muridae); Jansa et al. (2004, Muridae); Mercer & Roth (2003, Sciuridae). Relationships within murid subfamilies are from Jaarola et al. (2004, Arvicolinae); McKenna & Bell (1997, Gerbilinae); Watts and Baverstock (1995, Murinae); Lecompte et al. (2002, Zelotomys); Carleton (1980, Neotominae) and Smith & Patton (1999, Sigmodontinae).

***Xenarthra.*** Relationships are those proposed by Murphy et al. (2001a).

Where no molecular phylogenies were available, the resolution was maximized using taxonomic data by Wilson and Reeder (1993), McKenna and Bell (1997) and Nowak (1999).

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**Tree in Nexus format**

```
#NEXUS
BEGIN TREES;
  Title Brain-BMRTree;
  TRANSLATE
    1 Acionyx_jubatus,
    2 Ailurus_fulgens,
    3 Alopex_lagopus,
    4 Arctictis_binturong,
    5 Arctogalidia_trivirgata,
    6 Bassariscus_sumichrasti,
    7 Canis_latrans,
    8 Canis_mesomelas,
    9 Eira_barbara,
    10 Enhydra_lutris,
    11 Felis_concolor,
    12 Felis_pardalis,
    13 Felis_serval,
    14 Felis_rufus,
    15 Felis_yagouaroundi,
    16 Fennecus_zerda,
    17 Fossa_fossa,
    18 Galerella_sanguinea,
    19 Genetta_tigrina,
    20 Gulo_gulo,
    21 Hyaena_hyaena,
    22 Lutra_lutra,
    23 Lycaon_pictus,
    24 Martes_americana,
    25 Martes_martes,
    26 Meles_meles,
    27 Mustela_erminea,
    28 Mustela_frenata,
    29 Mustela_vison,
    30 Nandinia_binotata,
    31 Nasua_narica,
    32 Nasua_nasua,
    33 Panthera_leo,
    34 Panthera_onca,
    35 Panthera_tigris,
    36 Paradoxurus_hermaphroditus,
    37 Potos_flavus,
    38 Procyon_lotor,
    39 Proteles_cristatus,
    40 Spilogale_putorius,
    41 Suricata_suricatta,
    42 Taxidea_taxus,
    43 Ursus_ursinus,
    44 Vulpes_vulpes,
    45 Anoura_caudifer,
    46 Artibeus_jamaicensis,
    47 Artibeus_lituratus,
    48 Carollia_perspicillata,
    49 Chalinolobus_gouldi,
    50 Cynopterus_brachyotis,
    51 Desmodus_rotundus,
    52 Diaemus_youngi,
    53 Diphyllia_ecaudata,
    54 Dobsonia_inermis,
    55 Dobsonia_praedatrix,
    56 Eonycteris_spelaea,
    57 Eptesicus_fuscus,
    58 Eumops_perotis,
    59 Glossophaga_soricina,
    60 Hipposideros_galeritus,
    61 Koopmania_concolor,
    62 Leptonycteris_curasoae,
    63 Macroderma_gigas,
    64 Macroglossus_minimus,
    65 Megaloglossus_woermanni,
    66 Melonycteris_melanops,
    67 Miniopterus_australis,
    68 Miniopterus_schreibersi,
    69 Molossus_molossus,
    70 Mormoops_megalophylla,
    71 Myotis_lucifugus,
    72 Myotis_nigricans,
    73 Myotis_yumanensis,
    74 Natalus_tumidirostris,
    75 Noctilio_albiventris,
    76 Noctilio_leporinus,
    77 Nyctimene_albiventer,
    78 Nyctophilus_geoffroyi,
    79 Paranyctimene_raptor,
    80 Peropteryx_macrotis,
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81 Phyllostomus_discolor,
82 Phyllostomus_elongatus,
83 Phyllostomus_hastatus,
84 Pteronotus_davyi,
85 Pteronotus_parnellii,
86 Pteronotus_personatus,
87 Pteropus_giganteus,
88 Pteropus_hypomelanus,
89 Pteropus_poliocephalus,
90 Pteropus_scapulatus,
91 Pteropus_vampyrus,
92 Rhinonycteris_aurantius,
93 Rhinophylla_pumilio,
94 Rousettus_aegyptiacus,
95 Rousettus_amplexicaudatus,
96 Saccopterix_bilineata,
97 Sturnira_lilium,
98 Sturnira_tildae,
99 Syconycteris_australis,
100 Tonatia_bidens,
101 Uroderma_bilobatum,
102 Vampyressa_pusilla,
103 Heterohyrax_brucei,
104 Procavia_capensis,
105 Atelerix_albiventris,
106 Chrysochloris_asiatica,
107 Condylura_cristata,
108 Echinops_telfairi,
109 Erinaceus_europaeus,
110 Hemicentetes_semispinosus,
111 Hemiechinus_auritus,
112 Limnogale_mergulus,
113 Microgale_cowani,
114 Microgale_dobsoni,
115 Microgale_talazaci,
116 Neurotrichus_gibbsi,
117 Scalopus_aquaticus,
118 Scapanus_townsendii,
119 Setifer_setosus,
120 Tenrec_ecaudatus,
121 Elephantulus_intufi,
122 Elephantulus_myurus,
123 Alouatta_palliata,
124 Aotus_trivirgatus,
125 Callithrix_jacchus,
126 Cebuella_pygmaea,
127 Cercopithecus_mitis,
128 Erythrocebus_patas,
129 Galagoides_demidovii,
130 Otolemur_garnetti,
131 Papio_ursinus,
132 Perodicticus_potto,
133 Propithecus_verreauxi,
134 Saguinus_geoffroyi,
135 Saimiri_sciureus,
136 Aethomys_namaquensis,
137 Agouti_paca,
138 Ammospermophilus_leucurus,
139 Aplodontia_rufa,
140 Apodemus_flavicollis,
141 Apodemus_sylvaticus,
142 Arvicola_terrestris,
143 Bathyergus_suillus,
144 Cavia_porcellus,
145 Chaetodipus_baileyi,
146 Chaetodipus_hispidus,
147 Chaetodipus_penicillatus,
148 Chinchilla_lanigera,
149 Clethrionomys_gapperi,
150 Clethrionomys_glareolus,
151 Clethrionomys_rufocanus,
152 Clethrionomys_rutilus,
153 Cricetomys_gambianus,
154 Cricetus_cricetus,
155 Cryptomys_hottentotus,
156 Cynomys_ludovicianus,
157 Dasyprocta_leporina,
158 Dicrostonyx_groenlandicus,
159 Dipodomys_agilis,
160 Dipodomys_deserti,
161 Dipodomys_heermani,
162 Dipodomys_merriami,
163 Dipodomys_microps,
164 Dipodomys_ordii,
165 Dipus_sagitta,
166 Erethizon_dorsatum,
167 Funisciurus_pyrrhops,
168 Galea_musteloides,
169 Geocapromys_ingrahami,
170 Geomys_bursarius,
171 Georychus_capensis,
172 Gerbillurus_paeba,
173 Gerbillus_dasyurus,
174 Gerbillus_nanus,
175 Gerbillus_pyramidum,
176 Glaucomys_volans,
177 Heliophobius_argentocinereus,
178 Heliosciurus_rufobrachium,
179 Heterocephalus_glaber,
180 Heteromys_desmarestianus,
181 Hydrochaeris_hydrochaeris,
182 Hydromys_chrysogaster,
183 Hystrix_africaeaustralis,
184 Isthomys_pirrensis,

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185 Jaculus_jaculus,
186 Jaculus_orientalis,
187 Lagostomus_maximus,
188 Lemmus_lemmus,
189 Lemmus_sibiricus,
190 Liomys_salvini,
191 Malacothrix_typica,
192 Marmota_flaviventris,
193 Marmota_monax,
194 Mastomys_natalensis,
195 Megadontomys_thomasi,
196 Meriones_hurrinae,
197 Meriones_unguiculatus,
198 Mesocricetus_auratus,
199 Microdipodops_megacephalus,
200 Micromys_minutus,
201 Microtus_agrestis,
202 Microtus_arvalis,
203 Microtus_californicus,
204 Microtus_guentheri,
205 Microtus_longicaudus,
206 Microtus_montanus,
207 Microtus_ochrogaster,
208 Microtus_pennsylvanicus,
209 Microtus_pinetorum,
210 Microtus_townsendii,
211 Mus_minutoides,
212 Myoprocta_agouchy,
213 Myopus_schisticolor,
214 Myoxus_glis,
215 Mystromys_albicaudatus,
216 Nannospalax_ehrenbergi,
217 Nannospalax_leucodon,
218 Napaeozapus_insignus,
219 Neofiber_alleni,
220 Neotoma_albigula,
221 Neotoma_cinerea,
222 Neotoma_fuscipes,
223 Notomys_alexis,
224 Ochrotomys_nuttalli,
225 Oligoryzomys_longicaudatus,
226 Ondatra_zibethicus,
227 Onychomys_torridus,
228 Otomys_irroratus,
229 Otomys_unisulcatus,
230 Paraxerus_cepapi,
231 Parotomys_brantsii,
232 Pedetes_capensis,
233 Perognathus_flavus,
234 Perognathus_longimembris,
235 Peromyscus_boylii,
236 Peromyscus_californicus,
237 Peromyscus_crinitus,
238 Peromyscus_eremicus,
239 Peromyscus_gossypinus,
240 Peromyscus_leucopus,
241 Peromyscus_maniculatus,
242 Peromyscus_megalops,
243 Peromyscus_polionotus,
244 Peromyscus_sitkensis,
245 Peromyscus_truei,
246 Podomys_floridanus,
247 Proechimys_semispinosus,
248 Pseudomys_heermansburgensis,
249 Rattus_fuscipes,
250 Rattus_lutreolus,
251 Rattus_sordidus,
252 Reithrodontomys_megalotis,
253 Rhabdomys_pumilio,
254 Saccostomus_campestris,
255 Sciurus_aberti,
256 Sciurus_carolinensis,
257 Sigmodon_hispidus,
258 Spalacopus_cyanus,
259 Spermophilus_beecheyi,
260 Spermophilus_beldingi,
261 Spermophilus_lateralis,
262 Spermophilus_paryii,
263 Spermophilus_richardsoni,
264 Spermophilus_tereticaudatus,
265 Spermophilus_townsendii,
266 Spermophilus_tridecemlineatus,
267 Spermophilus_undulatus,
268 Stochomys_longicaudatus,
269 Tachyoryctes_splendens,
270 Tamias_amoenus,
271 Tamias_minimus,
272 Tamias_palmeri,
273 Tamias_striatus,
274 Tamiasciurus_hudsonicus,
275 Tatera_afra,
276 Tatera_indica,
277 Thomomys_bottae,
278 Thomomys_talpoides,
279 Xerus_inauris,
280 Zapus_hudsonicus,
281 Tupaia_glis,
282 Urogale_everetti,
283 Bradypus_variegatus,
284 Choleopus_hoffmanni,
285 Cyclopes_didactylus,
286 Dasypus_novemcinctus,
287 Euphractus_sexcinctus,
288 Tamandua_tetradactyla,

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289 Manis_javanica,
290 Manis_tricuspis,
291 Orycteropus_afer,
292 Papio_anubis,
293 Colobus_guereza,
294 Tarsius_spectrum,
295 Tarsius_syrichta,
296 Cheirogaleus_medius,
297 Eulemur_fulvus,
298 Galago_senegalensis,
299 Otolemur_crassicaudatus,
300 Loris_tardigradus,
301 Nycticebus_coucang,
302 Octodon_degus,
303 Geomys_pinetis,
304 Heteromys_anomalus,
305 Microtus_mexicanus,
306 Desmodillus_auricularis,
307 Acomys_cahirinus,
308 Conilurus_penicillatus,
309 Notomys_cervinus,
310 Thallomys_paedulcus,
311 Phyllotis_darwini,
312 Scotinomys_teguina,
313 Scotinomys_xerampelinus;

TREE Nee =
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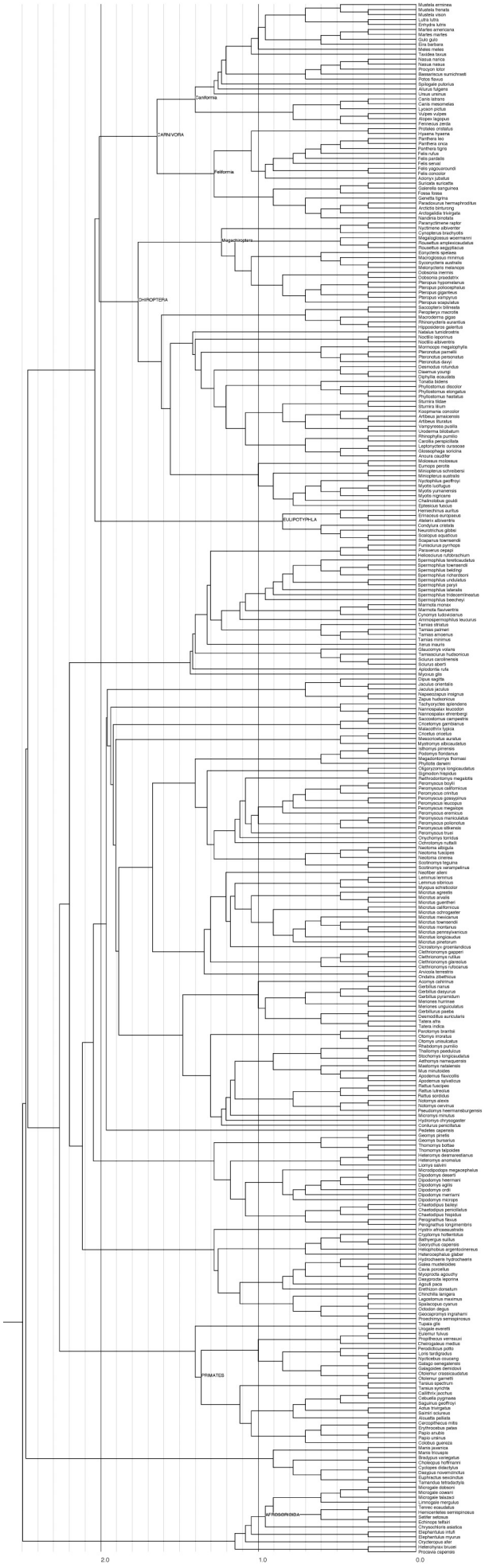
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END;

## Tree Figure

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Mustela erminea
Mustela frenata
Mustela vison
Lutra lutra
Enhydra lutris
Martes americana
Martes martes
Gulo gulo
Eira barbara
Meles meles
Taxidea taxus
Nasua narica
Nasua nasua
Procyon lotor
Ailurus fulgens ramchirasti
Ailurus fulgens
Spilogale putorius
Ursus ursinus
Canis latrans
Canis mesomelas
Lycaon pictus
Vulpes vulpes
Otocyon megalotis
Fennecus zerda
Proteles cristatus
Hyaena hyaena
Panthera leo
Panthera onca
Panthera tigris
Felis rufus
Felis pardalis
Felis wiedi
Felis yagouaroundi
Felis concolor
Acinonyx jubatus
Suricata suricatta
Galerella sanguinea
Fossa fossa
Genetta tigrina
Arctictis binturong
Arctogalidia trivirgata
Nandinia binotata
Paradoxurus hermaphroditus
Paguma larvata